

# HARSHAVARDHAN CHINTAPATLA

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## PROFESSIONAL SUMMARY

Software Engineer with 5 years of experience designing and delivering Java full-stack applications, RESTful backend services, AWS-backed cloud solutions, and AI-integrated platforms. Built production-style systems using Java, Spring Boot, Spring Security, Hibernate/JPA, AngularJS, React, MySQL, MongoDB, and AWS. Experienced in CI/CD with Jenkins, OAuth2, custom session-token security, and Spring Security filter design, controller-service-repository architecture, and responsible AI integration using RAG pipelines. Secondary strengths in enterprise-grade API validation, release lifecycle ownership, and test automation.

## WORK EXPERIENCE

### University of Missouri | Graduate Research Assistant, Software Engineering Jan 2024 - May 2026 | Columbia, MO

- Designed and built a two-tier Spring Boot application (MizzouCloudDevOps Learning Portal) separating the web tier (GitHub OAuth2 identity establishment, custom session token creation stored in active\_sessions, Thymeleaf shell, AngularJS routing) from the API tier (TokenAuthFilter validating X-Auth-Token before controller execution, @PreAuthorize RBAC, controller-service-repository layering, REST endpoints, MySQL/JPA persistence); implemented server-side assessment grading and PDF certificate generation via Apache PDFBox, serving 850+ students.
- Integrated a GPT-4 instructor assistant using a RAG pipeline: course content indexed into ChromaDB, instructor queries trigger top-k retrieval, and prompts are augmented with live SQL analytics from read-only MySQL analytics views; LLM tool calls are restricted to a controlled registry to prevent open-ended database access and guard against unsafe data exposure.
- Eliminated N+1 enrollment query patterns by replacing per-module database calls with a single bulk fetch and in-memory HashSet lookup; provisioned Amazon EC2 instances for browser-based cybersecurity lab environments and Amazon S3 for course asset and certificate storage; managed Jenkins CI/CD pipelines and enforced endpoint-level access controls around analytics and AI assistant APIs.

### Capgemini Technology Services | Associate Consultant Jun 2021 - Dec 2023 | Hyderabad, India

- Owned STLC execution and release validation for McDonald's NPOS POS modernization across four market deployments: authored manual Jira test cases, executed smoke, regression, SIT, and UAT cycles, and managed the defect lifecycle from discovery through triage, retest, and closure using Cypress, Postman/Newman, Jenkins, and Jira.
- Built a Cypress TypeScript UI automation suite covering POS transaction flows (login, menu browsing, item and combo selection, modifier application, promotion and discount logic, tax calculation, payment tendering, receipt validation, and order routing); validated REST service behavior across auth, menu, order lifecycle, promotion engine, tax, payment, and receipt APIs using Postman collections executed via Newman in Jenkins CI with JUnit XML reporting.
- Identified and documented release-blocking defects including payment completion failures, tax calculation mismatches, and receipt total errors; produced structured bug reports with reproduction steps, severity, priority, and evidence; contributed directly to go/no-go release decisions before market go-live across four deployments.

## PROJECTS

### MizzouCloudDevOps Learning Portal | Java 8, Spring Boot, Spring Security, Hibernate/JPA, AngularJS, Thymeleaf, MySQL, AWS, GPT-4 API, ChromaDB

- Built a two-tier Spring Boot application with a dedicated web tier (GitHub OAuth2, session token issuance, Thymeleaf index shell, AngularJS SPA routing) and a separate API tier (TokenAuthFilter validating X-Auth-Token on every request, @PreAuthorize RBAC, controller-service-repository layering, JPA/Hibernate persistence); graded assessments server-side to prevent client-side score manipulation and generated completion certificates using Apache PDFBox.
- Designed a GPT-4 instructor assistant with a controlled RAG pipeline: course content and lab materials indexed into ChromaDB, instructor queries trigger top-k semantic retrieval, and prompts are enriched with live student analytics from read-only MySQL views; a tool registry restricts LLM-callable operations to safe, scoped queries, preventing unrestricted access to the application database schema.
- Resolved N+1 query bottleneck in enrollment loading by replacing repeated per-module SELECT calls with a single bulk fetch and in-memory HashSet membership lookup; deployed browser-based cybersecurity lab environments on Amazon EC2 and stored course assets and generated PDF certificates in Amazon S3; maintained Jenkins CI/CD pipelines and enforced endpoint-level access controls around analytics and AI assistant APIs.

### RISE / PAALAB Research Chatbot | Node.js, React 18, Ollama (LLaMA 3.1 8B), Qdrant, nomic-embed-text, MongoDB, AWS S3/CEPH, Server-Sent Events

- Built a privacy-preserving local RAG pipeline running LLaMA 3.1 8B via Ollama on-premises, ensuring research data never leaves the university network; embedded domain knowledge (GSD tables, SEM manuals, crop guides) and live MongoDB records into Qdrant using 768-dimension nomic-embed-text vectors with group-level payload filtering to isolate RISE and PAALAB tenant contexts.
- Designed the query pipeline to parallelize live MongoDB context fetching (active jobs, recent results) with Qdrant vector retrieval per request; assembled an augmented prompt combining retrieved chunks, live data, and conversation history before streaming to LLaMA, enabling project-specific grounded responses without sending data to external cloud APIs.
- Implemented async Qdrant indexing triggered on job completion via setImmediate to avoid blocking the ingestion path; built a React 18 streaming UI using fetch() and ReadableStream over Server-Sent Events to render LLM tokens progressively for a responsive conversational experience.

### Remote Patient Health Monitoring Platform | AWS (API Gateway, Lambda, SQS, RDS, S3), Python, TypeScript, React

- Architected a serverless ingestion pipeline using AWS API Gateway and Lambda to validate IoT health payloads, archive raw readings in Amazon S3, and persist structured records in Amazon RDS; decoupled ingestion from downstream processing using Amazon SQS as a durable buffer so both sides scale and fail independently under variable device workloads.
- Implemented Python Lambda processing workers that normalize incoming readings, upsert structured records into RDS, aggregate per-patient metrics, apply threshold-based anomaly flagging, and write alert records for readings that fall outside configured vital-sign bounds.
- Built a React and TypeScript dashboard consuming processed health analytics from REST APIs exposed through API Gateway and a read-only Lambda query layer; the TypeScript frontend uses typed domain models and a typed API service layer that mirrors the backend contract for patient list, vital detail, and alert views.

- McDonald's NPOS Release Automation and API Validation** | *Cypress, TypeScript, Postman, Newman, Jenkins, GitHub, Jira*
- Automated the complete POS transaction flow using Cypress and TypeScript: login, menu browsing, item and combo selection, modifier application, promotion and discount logic, tax calculation, cash/card/contactless payment tendering, receipt validation, and order routing; executed via Jenkins CI with JUnit/HTML report publishing as release gate evidence.
  - Built Postman collections for all NPOS REST services including auth, menu retrieval, order lifecycle, promotion engine, tax calculation, payment processing, and receipt generation; executed via Newman in Jenkins CI pipelines and used report output to support go/no-go release decisions across four market deployments.
  - Authored Jira test cases and defect tickets across smoke, regression, and integration test types; led SIT validation for cross-service data consistency and supported UAT with business stakeholders to reach defect closure before each market go-live.
- OrangeHRM Java Automation Framework** | *Java, Selenium WebDriver, TestNG, Maven, Allure Reports, Jenkins, GitHub Webhooks, Jira*
- Built a Selenium Java framework using Page Object Model with BaseTest, ThreadLocal DriverFactory for parallel-safe driver management, explicit-wait-only WaitUtil (no Thread.sleep), and ScreenshotUtil capturing failure screenshots attached directly to Allure report steps; implemented TestNG DataProvider for data-driven login and PIM tests with smoke and regression suite separation.
  - Designed full STLC coverage on the public OrangeHRM demo portal: manual test cases linked to Jira stories via a requirement traceability matrix, defect classification distinguishing product bugs from locator, data, and environment failures, and smoke/regression/E2E suite tagging for targeted CI execution.
  - Configured a Jenkins declarative pipeline triggered by GitHub webhooks: checkout, headless Chrome test execution via Maven and TestNG, Allure and Surefire report publishing, screenshot archiving, and parameterized browser and suite selection to support fast smoke runs on every push and full regression on release candidates.

EDUCATION

**University of Missouri** | Master of Science, Computer Science Jan 2024 - May 2026

**Bharath Institute of Higher Education and Research** | Bachelor of Technology, Electronics and Communication Engineering Jul 2017 - Jun 2021

TECHNICAL SKILLS

|                |                                                                                                                          |
|----------------|--------------------------------------------------------------------------------------------------------------------------|
| Languages      | Java, Python, JavaScript, TypeScript, SQL, HTML5, CSS3                                                                   |
| Backend        | Spring Boot, Spring Security, Spring MVC, Spring Data JPA, Hibernate, REST APIs, OAuth2, Node.js, Express.js             |
| Frontend       | React, AngularJS, Thymeleaf, Chart.js, Bootstrap                                                                         |
| Cloud & DevOps | Amazon EC2, Amazon S3, Amazon RDS, AWS Lambda, AWS API Gateway, Amazon SQS, Jenkins, Docker, Git, GitHub Webhooks, Maven |
| Databases      | MySQL, MongoDB, Qdrant, ChromaDB                                                                                         |
| AI & LLMs      | GPT-4 API, LLaMA 3.1 via Ollama, RAG pipelines, vector embeddings, ChromaDB retrieval, LLM tool calling                  |
| Testing & QA   | Selenium WebDriver, Cypress, TestNG, Postman, Newman, Page Object Model, Allure Reports, Jira                            |
| Data & ETL     | ETL pipelines, RClone, S3-compatible CEPH storage, Apache PDFBox, Pandas                                                 |